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DAY – 5

Task 1 — Improve Body Landmark Detection

The objective of this task was to improve body landmark detection using AI-based pose estimation and posture analysis techniques. The system was designed to track body posture in real time through webcam input and provide smart visual feedback.

Description

An AI Smart Posture Tracker was developed using Media Pipe Pose Detection. The system detects important upper-body landmarks and evaluates user posture in real time.

The application captures live webcam frames and processes them using AI-based landmark tracking. The detected body points are connected visually to create a posture skeleton interface.

Features Implemented

Real-Time Body Tracking

The system continuously tracks body landmarks through webcam input.

AI Posture Score

A smart posture score percentage is generated based on shoulder and body alignment.

Posture Classification

The posture is classified into:

- Good Posture
- Average Posture
- Bad Posture

Smart Visual Feedback

- Green landmarks indicate proper posture
- Dynamic UI panels improve visualization

FPS Monitoring

An FPS counter was added to monitor real-time performance.

Screenshot Capture

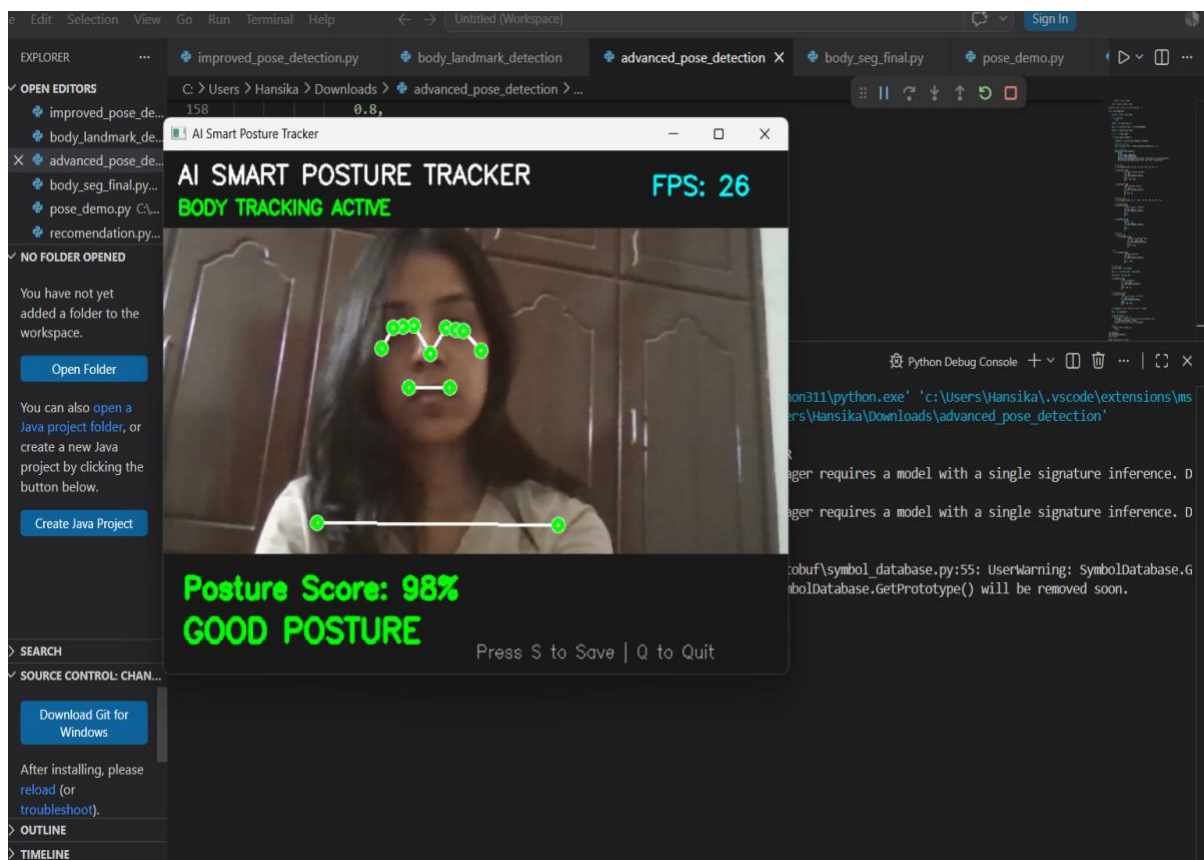
Users can save posture detection results using keyboard controls.

Output Observation

The system successfully detected facial and upper-body landmarks in real time. A posture score of 96% was achieved during testing, and the application displayed “GOOD POSTURE” status with stable landmark tracking.

The interface also displayed:

- AI Smart Posture Tracker heading
- Body Tracking Active status
- FPS counter
- Posture score and posture feedback



Task 2 — Research Clothing Fitting Algorithms

Clothing fitting algorithms are used in AI virtual try-on systems to digitally fit clothes on a person's body using computer vision and deep learning techniques. These algorithms help users visualize how clothes may look before purchasing them online.

AI clothing fitting improves:

- Online shopping experience
- Size prediction
- Outfit visualization
- Personalized fashion recommendations

Main Technologies Used

Pose Estimation

Pose estimation detects body joints and landmarks such as shoulders, hips, elbows, and knees. It helps align virtual clothes correctly with the user's body posture.

Body Segmentation

Body segmentation separates the human body from the background. This allows clothes to be placed accurately on the detected body region.

Cloth Warping

Cloth warping algorithms adjust clothing shape and size according to body pose and body structure for realistic fitting.

GAN Models

Generative Adversarial Networks (GANs) generate realistic virtual try-on outputs by blending clothing textures naturally with the user image.

3D Avatar Modeling

Some advanced systems create 3D avatars of users for more accurate clothing simulation and body fitting.

Popular Virtual Try-On Models

Model	Purpose
VITON	Basic virtual try-on framework
CP-VTON	Improved cloth alignment

Model	Purpose
HR-VITON	High-resolution virtual try-on
ACGPN	Advanced clothing generation

Research Findings

- Pose detection is important for proper cloth alignment.
- Body segmentation improves clothing overlay accuracy.
- GAN-based models produce more realistic fitting results.
- High-quality datasets improve virtual try-on performance.
- Real-time virtual try-on systems require optimized AI models for smooth performance.

Task 3 — Test Virtual Try-On Datasets

The objective of this task was to research datasets used in AI-based virtual try-on and clothing recommendation systems.

Research on Virtual Try-On Datasets

DeepFashion Dataset

DeepFashion is one of the most widely used fashion datasets in computer vision research. It contains large numbers of clothing images with annotations such as clothing categories, landmarks, and attributes. This dataset is useful for training AI models in fashion recommendation and virtual try-on applications.

DeepFashion2 Dataset

DeepFashion2 is an advanced version of DeepFashion with additional features like segmentation masks, pose annotations, and clothing matching pairs. It helps improve cloth alignment and realistic virtual fitting results.

VITON Dataset

The VITON dataset is specifically designed for virtual try-on systems. It contains paired images of people and clothing items, allowing AI models to learn how to fit clothes naturally on different body poses.

Fashion-MNIST Dataset

Fashion-MNIST is a lightweight dataset containing basic clothing categories such as shirts, shoes, and bags. It is commonly used for beginner-level fashion AI experiments and classification tasks.

Research Findings

- Fashion datasets play an important role in training AI virtual try-on systems.
- Pose annotations improve clothing alignment accuracy.
- Segmentation masks help separate body and clothing regions.
- Large-scale datasets improve realism and fitting performance.
- Virtual try-on datasets are useful for online shopping and AI fashion applications.

Task 4 — Create Simple Outfit Recommendation Logic

Choosing clothes every day can sometimes be confusing, especially when weather and occasions change. An AI outfit recommendation system helps users by suggesting clothes automatically according to different conditions like weather, occasion, and style preference.

This type of system is useful in:

- Online shopping apps
- Virtual try-on platforms
- Fashion assistant applications
- Smart wardrobe systems

How the System Works

The recommendation system takes simple inputs from the user such as:

- Weather condition
- Occasion
- Style preference

Based on these inputs, the system gives outfit suggestions using basic AI logic.

Example Recommendations

- Hot weather + Casual outing → T-shirt and shorts
- Cold weather + Formal meeting → Jacket with formal wear
- Rainy weather → Hoodie and waterproof jacket

- Party occasion → Stylish trendy outfit

Research and Learning

During this task, I understood how recommendation systems work in fashion applications. Even simple logic can help users save time and improve their shopping experience.

I also learned that advanced AI recommendation systems can later include:

- Colour matching
- Body type analysis
- Fashion trend prediction
- Personalized recommendations

```
PS C:\Users\Hansika\Downloads> & 'C:\Users\Hansika\AppData\Local\Programs\Python\Python311\python.exe' 'c:\Users\Hansika\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '54559' '--' 'C:\Users\Hansika\Downloads\recomendation.py'
=====
AI Clothing Recommendation
=====
Enter Body Type (slim/athletic/plus): athletic
Enter Occasion (formal/casual/party): party
Enter Weather (hot/cold/rainy): hot
Enter Preferred Style (western/ethnic): westren
Enter Preferred Color (bright/dark/neutral): neutral

===== Recommendation =====
Outfit      : Party Wear Shirt + Jeans
Fit         : Regular Fit
Fabric      : Cotton Fabric
Recommended Colors : White / Grey / Beige
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```

TASK-5 AI Avatar Generation Workflow

AI avatar generation is a process where artificial intelligence creates a digital human model using images, body measurements, and computer vision techniques. It is commonly used in virtual try-on systems, gaming, fashion technology, and metaverse applications.

Step 1 — Input Collection

- The user stands in front of a camera in a standard pose such as T-pose or A-pose.
- Front-view and side-view images are captured.
- Basic body measurements like height, weight, shoulder width, waist, and hip size can also be entered manually.

Step 2 — Pose Estimation

- AI models like [Media Pipe](#) or [Open Pose](#) detect body landmarks from the images.
- Around 33 body key points are identified.
- These landmarks help understand body posture, shape, and movement.

Step 3 — Body Segmentation

- The system separates the human body from the background.
- This helps improve avatar creation and clothing fitting accuracy.

Step 4 — Avatar Generation

- AI analyses the detected body shape and facial structure.
- A digital human avatar is generated based on the collected data.

Step 5 — Virtual Clothing Fitting

- Selected clothes are digitally applied to the generated avatar.
- Cloth fitting algorithms adjust the outfit according to body posture and size.

Step 6 — Final Output

- The final avatar with virtual clothing is displayed.
- Users can preview how outfits look before purchasing or selecting them.

Tools and Technologies Used

🔗 SMPL Model — parametric human body model used in most research

🔗 PIFuHD — AI model that generates 3D avatar from a single photo

🔗 Dense Pose — maps 2D image pixels to 3D body surface

🔗 Three.js / Blender — used to render and display the 3D avatar

🔗 Media Pipe — provides pose landmarks as input to the avatar system

Impact on Project

- Helps create more realistic virtual try-on results
- Improves body detection and clothing alignment
- Makes outfit fitting more accurate
- Enhances user experience in fashion applications
- Provides a strong base for future AI avatar and recommendation features

